KEYAN DING

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RESEARCH INTEREST

Empirical AI safety research, focusing on the evaluation of agentic systems, failure propagation, monitoring reliability, and safety assurance in complex deployments.

RESEARCH EXPERIENCE

Hallucination Propagation in LLM Reasoning (report) *Sep 2026 - Now*
Independent research | Python, PyTorch, LLM evaluation

- Designed **controlled LLM evaluations** to investigate intermediate error propagation in multi-step reasoning.
- Built a **reproducible evaluation framework** with counterfactual controls, validity checks, and automated verification.
- Identified **20 naturally generated errors** in a 120-case pilot, observing downstream error propagation under controlled state mappings.
- **Ongoing research:** Investigating how context contamination and intermediate-state manipulation affect error correction and downstream model behavior.

LLM Evaluation for Aviation Certification Reasoning (code) *Apr 2026 - Now*
Group research | RAG, LLM Evaluation

- Built a citation-grounded RAG system over FAA AC 21.17-4, using structured document chunking and embedding-based retrieval to enhance factuality in certification-related QA.
- Developing an **aviation evaluation benchmark** to measure answer faithfulness, citation precision, and abstention recall under ambiguous regulatory guidance.

Enzyme Stability Prediction using Protein Language Models (code) *Dec 2025*
Independent research | PyTorch, Fine-tuning

- **Fine-tuned** a 420M-parameter ProtBERT model for enzyme melting-temperature prediction, implementing layer freezing and cosine learning-rate scheduling to improve training efficiency.
- Built a **reproducible PyTorch pipeline** with 5-fold cross-validation and model ensembling to improve generalization across protein sequences while reducing overfitting.

PROFESSIONAL EXPERIENCE

Systems Engineer — Safety-Critical Flight Guidance System *Sep 2023 - Now*
Commercial Aircraft Corporation of China, Ltd.

- Led FMS guidance systems engineering, translating operational and safety objectives into traceable requirements and verification criteria.
- Developed an **automated MATLAB/Simulink verification pipeline** for scenario-based testing, requirements coverage, and regression analysis.
- Designed a **control-law tuning workflow** using representative operating points and system identification to characterize inner-loop dynamics.
- Proposed an **evidence-based gain-tuning strategy**, adapting controller gains to nominal and critical operating conditions to mitigate stability risks.
- Developing **experimental performance criteria** through iterative simulation and tuning, targeting consistent tracking and stability with margins for model uncertainty

RESEARCH EXPERIENCE (CONTINUED)

Value Approximation in Offline Reinforcement Learning (pdf) *Aug 2022 - Dec 2022*
course Research / RL Theory

· Analyzed offline RL algorithms (FQI, MSBO, SBEED), comparing function-approximation assumptions, convergence guarantees, and sample complexity.

Geometric Optimization and Covering Problems (site) *Jan 2022 - Aug 2022*
Graduate Mentor, Illinois Geometry Lab

· Studied an open-ended geometric question: the maximum area coverable by finitely many triangles in a disc using constrained optimization and asymptotic analysis. Derived precise upper and lower bounds on approximation error.

· The work resulted in a peer-reviewed publication in *The PUMP Journal of Undergraduate Research (2025).*

EDUCATION BACKGROUND

University of Illinois at Urbana-Champaign *Aug 2021 - May 2023*
M.S. in Mathematics *GPA:* 3.5/4.0

· Selected coursework: Statistical Reinforcement Learning, Pattern Recognition, Online Learning, Theory of Probability, Mathematical Statistics

University of Washington, Seattle *Sep 2017 - Jun 2021*
B.S. in Mathematics *GPA:* 3.6/4.0

· Selected coursework: Machine Learning, Data Structures and Algorithms, Mathematical Modeling, Linear Algebra, Topology, Stochastic Processes

PUBLICATION

Covering Area in a Disc by Finitely Many Triangles. (pdf)

Chen, Z., **Ding, K.**, Huang, Y., Jia, X., Rosenblatt, J., Su, Y., & Xue, J. (2025). In *The PUMP Journal of Undergraduate Research*, 8, 47–56.

TECHNICAL SKILL

Programming	Python, PyTorch, Hugging Face, MATLAB/Simulink, C/C++, SQL
ML Methods	Transformer fine-tuning, reinforcement learning, LLM inference & evaluation, RAG
Evaluation & Research	Experimental design, counterfactual testing, statistical analysis, scenario-based testing, reproducible-workflow development, failure-mode analysis
Tools	Git, GitHub, LaTeX, Linux
Speaking Languages	Mandarin(Native), English(Fluent)